

Searching 2D Lists

Look at this code:

```
clubs = [["art", 12], ["chess", 18], ["code", 15]]
target = input("Club to find: ")
target = target.lower()
found = False
for club in clubs:
    if club[0] == target:
        print("Members:", club[1])
        found = True
if found == False:
    print("Club not found.")
```

Use the code above to help you. After completing each challenge below, run your program to test that it works correctly. Save each completed challenge program in a new file.

Programming Challenges

Challenge 1

Change the example so it searches by member count. Ask for a whole number in `target` with `int()` and remove the `lower()` line. Compare `target` with each row's count and display a match after `"Club:"`. For example, 15 displays `"Club: code"` and 99 displays `"Club not found."`

Challenge 2

Change the example so it updates the club it finds. When a name matches, ask for `new_members`, replace that row's count and display the row after `"Updated:"`. For example, `"Chess"` and 20 display `"Updated: ['chess', 20]"`; `"Music"` displays `"Club not found."`

Challenge 3

Create a cafe order program with a 2D list called `menu`: `["tea", 2]`, `["cake", 3]` and `["soup", 5]`. Ask for an item, use `lower()` on it, then ask for a whole-number quantity. Use a `for` loop to find the item and display `"Cost:"` with its price times the quantity. If nothing matches, display `"Item not found."` after the loop.

